MALVIKA JADHAV

malvikajadhav.github.io | malvikajadhav99999@gmail.com | linkedin.com/in/malvika-jadhav

PhD researcher in LLM safety and model internals: causal interventions on open-weight models, large-scale misuse measurement, and safety benchmarking against production guardrails.

EDUCATION

University of Florida, PhD, Computer & Information Science & Engineering **Expected May 2028**
GPA 3.82 · Advisor: Dr. Vincent Bindschaedler · Florida Institute for Cybersecurity Research

University of Florida, MS, Computer & Information Science & Engineering
Coursework: Machine Learning, Applied ML, Trustworthy ML, Distributed Systems

University of Pune, BE, Computer Engineering

EXPERIENCE

Graduate Research Assistant,University of Florida **Aug 2024 - Present**

GenAI misuse measurement: the synthetic NCII creation ecosystem

- Built a hybrid NLP pipeline (SetFit few shot classification, knowledge graph construction) over 280K+ 4chan posts and 78K+ Reddit submissions, producing the first platform scale map of how synthetic intimate image abuse gets produced, shared, and sold.
- Identified the standardized misuse workflows behind it, including base model selection and LoRA adapters, plus the commercial nudification services and referral networks that keep the ecosystem running, giving platforms concrete signals for detection, enforcement, and content policy.

Model internals for safer code: eliciting latent security knowledge in LLMs

- Localized secure coding behavior inside code LLMs to specific attention heads and MLP neurons; found 40.5% of security components also carry general coding ability, explaining why blunt steering trades capability for safety.
- Built a training free component editing method that targets security specific components only: cut vulnerable code rates by up to 34.9 points on ProSec and CyberSecEval across CodeLlama 7B, Qwen2.5-Coder 3B, and Qwen3 14B while raising pass@1 by up to 6.5 points, beating a fine tuned baseline with no training or inference cost.

Interpretability of bias in LLM safety advice

- Building an evaluation and intervention pipeline (PyTorch, HuggingFace, nnsight) over 480 real narratives with controlled counterfactuals across gender and eight inferred attributes (dependence, entitlement, credibility, culpability, urgency, and more), scored on a categorical action space that exposes shifts scalar bias metrics miss.
- Mapping how open-weight LLMs (Qwen 3.5 9B/27B, Gemma 4) infer a help-seeker's standing and how it gates advice, via linear probes, activation steering, ablation and causal mediation on B200s; testing cross-model and clinical-domain transfer and whether production guardrails share the bias.

Inclusive model evaluation: deepfake detection across hearing abilities

- Co-designed and ran an in person study with 80 participants (2,400 trial level responses, mixed effects logistic regression) measuring how deepfake susceptibility shifts across hearing profiles, a template for evaluating AI systems by subgroup rather than in aggregate.

Abuse measurement at scale: Android stalkerware

- Engineered a pipeline combining static, dynamic, and taint analysis with multilingual NLP (NLLB, 43 languages) across 8,421 apps and 9 Android versions, showing that 98.6% of stalkerware apps misrepresent what they do compared with 14% of benign apps.

PUBLICATIONS & INVITED TALKS

- *It Takes a Forum: Resources & Stakeholders Sustaining the Online Synthetic NCII Creation Ecosystem.* ACM CCS 2026
- *Uncovering the Deceptive Tactics of Stalkerware: A Large-Scale Measurement Study.* ACM ASIACCS 2026
- *"What I See is What I Hear": Deepfake Detection Across Diverse Hearing Abilities.* NDSS 2027.
- PolicyEngine: A Defensive Framework Against Cross-Channel Location Validation in Android Stalkerware. ACSAC 2026.
- *Inference Attacks for X-Vector Speaker Anonymization.* IEEE Security & Privacy Workshops, 2025.
- *Thwart Me If You Can: An Empirical Analysis of Android Platform Armoring Against Stalkerware.* Preprint, arXiv:2508.02454
- *"When Your Threat Model Lives With You: How Stalkerware Breaks Traditional Security Paradigms."* Weaponized Software panel, Tech Abuse Conference, UCL Gender & Tech Research Lab, London, May 2026.

SKILLS

ML engineering: PyTorch, HuggingFace Transformers, nnsight, TensorFlow, scikit-learn, SetFit, LLM fine-tuning, activation steering and model editing, linear probing, distributed training, mixed-effects modeling

Responsible AI and safety: red teaming, safety benchmark design, guardrail evaluation (Llama Guard 4, ShieldGemma, OpenAI Moderation, Perspective API), misuse measurement, subgroup evaluation, annotation framework design

Systems: Python, C++, Git, Linux, static/dynamic/taint analysis (JADX, FlowDroid), CUDA